

PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DARES SALAAM REGION

FORM SIX MOCK EXAMINATION

CHEMISTRY 3B

Code: 132/3B

Time: 3.00 Hours

Tuesday 20th February 2024 A.M.

INSTRUCTIONS

1. This paper consists of **three (03)** questions. Answer **all** questions.
 2. Question one carries **20 marks** and the other two carries **15 marks** each.
 3. All answers must be written in the answer booklets provided.
 4. Mathematical tables, and no – programmable calculators may be used.
 5. Cellular phones and any other unauthorized materials are **not** allowed in the examination room.
 6. Write your examination number on every page of your answer booklet(s)
- Atomic masses
 $H = 1, O = 16, Na = 23, Cl = 35.5, C = 12, Mn = 55, K = 39, Cu = 63.5$
■ Density of water = $1g/cm^3$
■ Specifics heat capacity water = $4.18J/g^{\circ}C$

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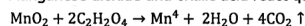
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This paper consists of 3 printed pages

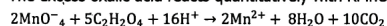
1. You are provided with the following solutions
 AA: A solution containing 1g of an impure sample of MnO_2 dissolve in 250cm^3 of distilled water and 2.2g of oxalic acid crystals ($\text{C}_2\text{H}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$)
 BB: A standard solution of 0.02M KMnO_4
 CC: 2M H_2SO_4 (Sulphuric acid)

Theory

Manganese dioxide and oxalic acid react quantitatively as follows



The excess oxalic acid reacts quantitatively with KMnO_4 . As follows



Procedures

- Pipette 20cm^3 (or 25cm^3) of AA into a conical flask
- Add about 20cm^3 of dilute H_2SO_4 and heat the content of flask to about 80°C
- Titrate the hot AA with BB in the burette until the purple colour no longer disappear
- Repeat this produce two times and record your results in a tabular form as shown below.

Results

The volume of the pipette used _____ cm^3

Burette reading

Titration number	Pilot	1	2
Final reading (cm^3)			
Initial reading (cm^3)			
Title volume (cm^3)			

Summary

For complete reaction _____ cm^3 of AA required _____ cm^3 BB

Calculate

- Molarity of oxalic acid
- Concentration of both hydrated and unhydrated oxalic acid in g/dm^3
- The percentage purity of the manganese dioxide in the sample.

2. You are provided with the following
 A_1 : 5g of hydrated ammonium oxalate $(\text{COONH}_4)_2 \cdot \text{H}_2\text{O}$
 A_2 : 5g of anhydrous sodium carbonate
 A_3 : Distilled water
 : Stirrer / glass rod, thermometer.

Procedure:

- Record the room temperature
- Pour 50cm^3 of A_3 into a 100ml clean beaker
- Record the temperature Tw .
- Add 5g of A_2 into a beaker containing A_3 . Stir gently until the dissolution is complete record the temperature Ts .
- Clean a beaker and the thermometer then repeat the experiment using 5g of A_1 and record Tw and Ts as in (iii) and (iv) above
- Tabulate your result as shown below

Substance	V of water	Mass of salt	Tw: $^\circ\text{C}$	Ts: $^\circ\text{C}$	ΔT
Na_2CO_3					
$(\text{COONH}_4)_2 \cdot \text{H}_2\text{O}$					

Questions.

- Calculate the molar mass of each salt
- Calculate the enthalpy change of solution of each salt
- Calculate the molar enthalpy change of solution of each salt
- State whether the process of dissolving each salt is endothermic or exothermic.
 Given that the specific heat capacity of water = $4.18\text{J/g}^\circ\text{C}$.

3. Sample "A" is a sample salt containing one cation and one anion. Carefully carry out qualitative analysis experiment to identify ions present in the salt based on the following test.

- Appearance of the sample
- Action of heat on sample
- Flame test on a sample
- Action of dil. HCl on a solid sample
- Solubility (divided the resulting solution into two portions)
- To the first portion add few drops of NaOH solution. Then add in excess.
- To the second portion add K_1 solution warm and cool the mixture.
- Transfer a small amount of the original sample into a test tube. Add copper turnings followed by concentrated sulphuric acid then warm.

Questions.

- Prepare a relevant table showing the qualitative analysis result.
- Write cation and anion present
- Write the formula of salt A.
- Write a balanced chemical equation for the reaction in experiment vi, vii

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(b) Concentration of hydrated oxalic acid.

From;

$$\text{Molarity} = \frac{\text{mass concentration}}{\text{Molar mass}} \quad (0.5 \text{ mark})$$

$$\text{Mass Conc.} = \text{Molarity} \times \text{Molar mass}$$

$$\text{But, Molar mass of } \text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O} = 126 \text{ g/mol}$$

$$\begin{aligned} \text{Mass Conc.} &= 0.02875 \text{ mol dm}^{-3} \times 126 \text{ g mol}^{-1} \quad (0.5 \text{ mark}) \\ &= 3.6225 \text{ g dm}^{-3} \quad (0.1 \text{ mark}) \end{aligned}$$

Concentration of anhydrous oxalic acid.

$$= 0.02875 \text{ mol dm}^{-3} \times 90 \text{ g mol}^{-1}$$

$$= 2.5875 \text{ g dm}^{-3} \quad (0.1 \text{ mark})$$

(c) Percentage purity of MnO_2 in the sample.

From;

$$\% \text{ purity} = \frac{\text{Conc. of pure}}{\text{Conc. of impure}} \times 100 \quad (0.1 \text{ mark})$$

$$\begin{aligned} \text{Concentration of impure } \text{MnO}_2 &= \frac{\text{mass}}{\text{Volume}} \\ &= \frac{1 \text{ g}}{0.25 \text{ dm}^3} \\ &= 4 \text{ g/dm}^3 \\ &\quad (0.1 \text{ mark}) \end{aligned}$$

MnO_2 was finished at the end of reaction.

Thus; $a - x = 0$ and $x = 0.021$

Then; $a - 0.021 = 0$

$$a = 0.021 \text{ M} \quad (01 \text{ mark})$$

Molar mass of $\text{MnO}_2 = 87 \text{ g/mol}$

Concentration (g/dm^3) = Molarity $\times$ Molar mass

$$= 0.021 \text{ mol/dm}^3 \times 87 \text{ g/mol}^{-1}$$

$$= 1.83 \text{ g/dm}^3$$

Concentration of pure $\text{MnO}_2 = 1.83 \text{ g/dm}^3$ (01 mark)

Then;

$$\% \text{ purity} = \frac{1.83 \text{ g/dm}^3}{4 \text{ g/dm}^3} \times 100$$

$$= 45.8\%$$

$\therefore$ Percentage purity of MnO_2 in the Sample = 45.8% (01 mark)

Enthalpy change for $(\text{COONH}_4)_2 \cdot \text{H}_2\text{O}$

From; $\Delta H = -mc\Delta T$

But; $\rho = \frac{\text{mass}}{\text{Volume}}$

Given; $\rho_{\text{H}_2\text{O}} = 1\text{g/cm}^3$ and $V_{\text{H}_2\text{O}} = 50\text{cm}^3$

$$\begin{aligned}\text{So, Mass} &= \rho_{\text{H}_2\text{O}} \times V_{\text{H}_2\text{O}} \\ &= 1\text{g/cm}^3 \times 50\text{cm}^3 \\ &= 50\text{g}\end{aligned}$$

$$\begin{aligned}\text{Then; } \Delta H &= -50\text{g} \times 4.18\text{J/g}^\circ\text{C} \times -2^\circ\text{C} \\ &= +418\text{J}\end{aligned}$$

$\therefore$ Enthalpy change for $(\text{COONH}_4)_2 \cdot \text{H}_2\text{O} = +418\text{J}$ (0.5 mark)

(c) Molar enthalpy change for each salt;
For Na_2CO_3 ;

$$q = \frac{\Delta H}{n} \quad (0.1 \text{ mark})$$

$$\text{But; } n = \frac{\text{mass}}{\text{Molar mass}} = \frac{5\text{g}}{106\text{g/mol}} = 0.047\text{mol} \quad (0.1 \text{ mark})$$

$$\begin{aligned}q &= \frac{-836\text{J}}{0.047\text{mol}} \\ &= -17787.234\text{Jmol}^{-1} \quad (0.2 \text{ mark})\end{aligned}$$

$\therefore$ Molar enthalpy change for $\text{Na}_2\text{CO}_3 = -17787.234\text{Jmol}^{-1}$

3. S/N	Experiment	Observation	Inference
(i)	Appearance of solid sample.	Salt A was a white crystalline solid (01 mark)	Non-transition metals may be present (00½ mark) NO_3^- , SO_4^{2-} , Cl^- , CO_3^{2-} , CrO_4^{2-} , NO_2^- , CH_3COO^- , $\text{Cr}_2\text{O}_7^{2-}$ may be present (00½ mark)
(ii)	Action of heat	Brown fumes evolved which turned moist blue litmus paper red. (00½ mark)	NO_3^- may be present except those of K^+ , Na^+ and NH_4^+ (00½ mark)
(iii)	Flame test	Salt A burns with a blue flame (00½ mark)	Pb^{2+} , Sb^{2+} may be present (00½ mark)
(iv)	Action of dil. HCl	No gas evolved. (00½ mark)	SO_4^{2-} , Cl^- , NO_3^- may be present (00½ mark)
(v)	Solubility	Salt A was soluble in cold water (00½ mark)	Soluble salt was present (00½ mark)
(vi)	Addition of NaOH in small amount then in excess.	In small amount of NaOH, white precipitates formed. In excess NaOH the precipitates dissolved. (00½ mark)	Sn^{2+} , Pb^{2+} may be present. (00½ mark)
(vii)	Addition of KI solution then heated.	Yellow ppt. formed which disappeared on warming but reappeared on cooling. (00½ mark)	Pb^{2+} Confirmed (00½ mark)
(viii)	Action of Copper turning to a solid sample followed by conc. H_2SO_4 then heated.	Brown fumes were evolved (00½ mark)	NO_3^- Confirmed. (00½ mark)